



PROGRAMMING FUNDAMENTALS

CS – 101

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EXERCISES

1. Using a dictionary as argument, sort and print out the key-value pairs, as sorted by their keys
2. Using a dictionary as argument, sort and print out the key-value pairs, as sorted by their values
3. Find min and max key in the above mentioned dictionary
4. Find min and max value in the above mentioned dictionary
5. Given a dictionary as argument, find the sum of all its values

EXERCISE # 1 SOLUTION

1. Using a dictionary as argument, sort and print out the key-value pairs, as sorted by their keys

- **def sort_by_key(d):**

- d1 = {}**

- for i in sorted(d):**

- d1[i] = d[i]**

- return d1**

- **h = {'b': 1, 'r': 2, 'o': 2, 'n': 1, 't': 1, 's': 2, 'a': 1, 'u': 2}**

- k1 = sort_by_key(h)**

- print(k1)**

EXERCISE # 2 SOLUTION

2. Using a dictionary as argument, sort and print out the key-value pairs, as sorted by their values

- ```
def sort_by_val(d):
 d1 = {}
 val = list(d.values())
 val.sort()

 for i in val:
 if(i not in d1.values()):
 for j in d:
 if (i == d[j]):
 d1[j] = d[j]

 return d1
```
- ```
h = {'b': 1, 'r': 2, 'o': 2, 'n': 1, 't': 1, 's': 2, 'a': 1, 'u': 2}  
k2 = sort_by_val(h)  
print(k2)
```

EXERCISE # 3 SOLUTION

3. Find min and max key in the above mentioned dictionary

- **def find_min_max_key(d):**
 minimum = 'z'
 maximum = 'a'
 for i in d:
 if(i < minimum):
 minimum = i
 if(i > maximum):
 maximum = i
 print(minimum, maximum)
- **h = {'b': 1, 'r': 2, 'o': 2, 'n': 1, 't': 1, 's': 2, 'a': 1, 'u': 2}**
 find_min_max_key(h)

EXERCISE # 4 SOLUTION

4. Find min and max value in the above mentioned dictionary

- ```
def find_min_max_val(d):
 d_vals = list(d.values())
 minimum = 999
 maximum = -1
 for i in d_vals:
 if(i < minimum):
 minimum = i
 if(i > maximum):
 maximum = i
 print(minimum, maximum)
```
- ```
h = {'b': 1, 'r': 2, 'o': 2, 'n': 1, 't': 1, 's': 2, 'a': 1, 'u': 2}  
find_min_max_val(h)
```

EXERCISE # 5 SOLUTION

5. Given a dictionary as argument, find the sum of all its values

- **def sum_of_vals(d):**
 sum = 0
 for i in d:
 sum = sum + d[i]
 return sum
- **h = {'b': 1, 'r': 2, 'o': 2, 'n': 1, 't': 1, 's': 2, 'a': 1, 'u': 2}**
 s = sum_of_vals(h)
 print(s)